G. Leigh Harwell

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Academic Website
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Education

University of Oklahoma August 2023 – May 2027
Bachelor of Science in Astrophysics *Norman, OK, USA*
- Minor: Mathematics | Certificate: Data Analytics
- GPA: 3.63

Research Experience

Multi-Messenger Observations of Compact Objects August, 2026 – Present
University of Oklahoma, Department of Physics and Astronomy *Norman, OK, USA*
- Characterizing double white dwarf systems through multi messenger observations.
- Analyze multi messenger datasets to identify compact binary candidates for follow-up observational studies.

Machine Learning and Stellar Parameter Estimations October, 2024 – October 2026
University of Oklahoma, Department of Physics and Astronomy *Norman, OK, USA*
- Transitioned project into an intensive NSF REU at the University of Oklahoma during Summer 2026, focusing on advanced stellar parameter estimation.
- Apply machine learning techniques to asteroseismology and galactic archaeology datasets to improve stellar classification models.
- Implementing "TheCannon" data driven model to estimate stellar parameters in Astroseismolog with elemental abundances from spectroscopic data.

Research Presentations

Stellar Label Estimate through use of Machine Learning October, 2026
Department Research Symposium, University of Oklahoma *Norman, OK, USA*
- **Harwell, G. L.**, Beaty, M., Hayden
- Poster presentation detailing the findings from the Provost's UReCA Summer Fellowship program over my research on machine learning and stellar parameter estimations.

Stellar Label Estimates Through Use of Machine Learning August, 2026
University of Oklahoma's NSF REU *Norman, OK, USA*
- **Harwell, G. L.**, Beaty, M., Hayden
- Fifteen minute final talk over our work during the summer and our findings.

Awards & Honors

Provost's UReCA Summer Fellowship 2026
The University of Oklahoma *Norman, OK, USA*
- Funded collaborative research developing machine learning tools for stellar label estimation; presented findings at department symposium.
- This allowed me to take part in the events put on by OU's NSF REU.

Physics and Astronomy Meritorious Scholarship 2024 – Present
University of Oklahoma, Homer L. Dodge Department *Norman, OK, USA*
- Awarded to outstanding physics and astronomy students that maintain high academic standing.

President's Honor Roll 2024 – Present
University of Oklahoma *Norman, OK, USA*
- Awarded to distinguished students that maintain a 4.0 GPA throughout the semester.

Other Experience

Physics for Life Science Majors Tutoring Coordinator & PLA August, 2025 – Present
University of Oklahoma, Student Learning Center, Action Tutoring *Norman, OK, USA*

- Coordinating the tutoring for the life science majors, and collaborating with department faculty to provide extra problem sets and conceptual worksheets for exam review.
- Provide individual and small-group tutoring sessions in introductory mechanics, optics, thermodynamics, and electromagnetism.

Specialized Skills

Programming Languages: Python (intermediate), LaTeX (beginner)
Data Analysis: Multi-messenger data analysis, stellar parameter modeling, and machine learning workflow.
Computer Literacy: Linux command-line operations, Git/GitHub version control, Jupyter Notebooks, remote server/cluster access (SSH), and automated data processing pipelines.
Adaptability: Respond flexibly to evolving factors, environments, and conditions by using new skills.

Extracurricular

Community and Belonging Committee| August 2026 – Present
Society of Physics Students| August 2023 – Present
Women in Physics| August 2023 – Present